

```

MySQL 8.0 Command Line Client
mysql> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| nit |
| performance_schema |
| sys |
+-----+
5 rows in set (0.05 sec)

mysql> use nit;
Database changed
mysql> create table student(name varchar(30),id int not null primary key, address varchar(50), marks int);
Query OK, 0 rows affected (0.06 sec)

mysql> desc student;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| name | varchar(30) | YES | | NULL | |
| id | int | NO | PRI | NULL | |
| address | varchar(50) | YES | | NULL | |
| marks | int | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.02 sec)

mysql> insert into student values('Sam',21,'hyd',73),('Dev',21,'goa',75),('Pavan',22,'pune',77),('Kalyan',28,'kolkata',78),('Arjun',23,'ongole',78),('Sriram',24,'warangal',79),('Aniket',23,'jammu',70);
ERROR 1062 (23000): Duplicate entry '21' for key 'student.PRIMARY'
mysql> insert into student values('Sam',21,'hyd',73),('Dev',19,'goa',75),('Pavan',22,'pune',77),('Kalyan',28,'kolkata',78),('Arjun',23,'ongole',71),('Sriram',24,'warangal',79),('Aniket',23,'jammu',70);
ERROR 1062 (23000): Duplicate entry '23' for key 'student.PRIMARY'
mysql> insert into student values('Sam',21,'hyd',73),('Dev',19,'goa',75),('Pavan',22,'pune',77),('Kalyan',28,'kolkata',78),('Arjun',23,'ongole',71),('Sriram',24,'warangal',79),('Aniket',25,'jammu',70);
Query OK, 7 rows affected (0.01 sec)
Records: 7 Duplicates: 0 Warnings: 0

mysql> select * from student;
+-----+-----+-----+-----+
| name | id | address | marks |
+-----+-----+-----+-----+
| Dev | 19 | goa | 75 |
| Kalyan | 28 | kolkata | 78 |
| Sam | 21 | hyd | 73 |
| Pavan | 22 | pune | 77 |
+-----+-----+-----+-----+

```

```
MySQL 8.0 Command Line Client
mysql> insert into student values('Sam',21,'hyd',73),('Dev',19,'goa',75),('Pavan',22,'pune',77),('Kalyan',28,'kolkata',78),('Arjun',23,'ongole',71),('Sriram',24,'warangal',79),('Aniket',25,'jammu',78);
Query OK, 7 rows affected (0.01 sec)
Records: 7  Duplicates: 0  Warnings: 0

mysql> select * from student;
+----+-----+-----+-----+
| name | id | address | marks |
+----+-----+-----+-----+
| Dev | 19 | goa | 75 |
| Kalyan | 20 | kolkata | 78 |
| Sam | 21 | hyd | 73 |
| Pavan | 22 | pune | 77 |
| Arjun | 23 | ongole | 71 |
| Sriram | 24 | warangal | 79 |
| Aniket | 25 | jammu | 78 |
+----+-----+-----+-----+
7 rows in set (0.00 sec)

mysql> select name from student;
+-----+
| name |
+-----+
| Dev |
| Kalyan |
| Sam |
| Pavan |
| Arjun |
| Sriram |
| Aniket |
+-----+
7 rows in set (0.00 sec)

mysql> select name,id from student;
+-----+-----+
| name | id |
+-----+-----+
| Dev | 19 |
| Kalyan | 20 |
| Sam | 21 |
| Pavan | 22 |
| Arjun | 23 |
| Sriram | 24 |
| Aniket | 25 |
+-----+-----+
7 rows in set (0.00 sec)
```

7 rows in set (0.00 sec)

mysql> select * from student where id=21;

name	id	address	marks
San	21	hyd	73

1 row in set (0.00 sec)

mysql> select * from student;

name	id	address	marks
Dev	19	goa	75
Kalyan	20	kolkata	78
San	21	hyd	73
Pavan	22	pune	77
Arjun	23	angola	71
Sriram	24	warangal	79
Aniket	25	jammu	70

7 rows in set (0.00 sec)

mysql> update student set address='coorg' where id=24;

Query OK, 1 row affected (0.01 sec)

Rows matched: 1 Changed: 1 Warnings: 0

mysql> select * from student;

name	id	address	marks
Dev	19	goa	75
Kalyan	20	kolkata	78
San	21	hyd	73
Pavan	22	pune	77
Arjun	23	angola	71
Sriram	24	coorg	79
Aniket	25	jammu	70

7 rows in set (0.00 sec)

mysql> alter table student add phoneNo int;

Query OK, 0 rows affected (0.03 sec)

Records: 0 Duplicates: 0 Warnings: 0

```

MySQL 8.0 Command Line Client
+-----+
| Aniket | 25 | jansu | 70 |
+-----+
7 rows in set (0.00 sec)

mysql> alter table student add phoneNo int;
Query OK, 0 rows affected (0.03 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> select * from student;
+-----+
| name | id | address | marks | phoneNo |
+-----+
| Dev | 19 | goa | 75 | NULL |
| Kalyan | 20 | kolkata | 78 | NULL |
| San | 21 | hyd | 73 | NULL |
| Pavan | 22 | pune | 77 | NULL |
| Arjun | 23 | ongole | 71 | NULL |
| Sriraa | 24 | coorg | 79 | NULL |
| Aniket | 25 | jansu | 70 | NULL |
+-----+
7 rows in set (0.00 sec)

mysql> update student set phoneNo=++;
ERROR 1864 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '++' at line 1
mysql> update student set phoneNo=000;
Query OK, 7 rows affected (0.01 sec)
Rows matched: 7 Changed: 7 Warnings: 0

mysql> select * from student;
+-----+
| name | id | address | marks | phoneNo |
+-----+
| Dev | 19 | goa | 75 | 0 |
| Kalyan | 20 | kolkata | 78 | 0 |
| San | 21 | hyd | 73 | 0 |
| Pavan | 22 | pune | 77 | 0 |
| Arjun | 23 | ongole | 71 | 0 |
| Sriraa | 24 | coorg | 79 | 0 |
| Aniket | 25 | jansu | 70 | 0 |
+-----+
7 rows in set (0.00 sec)

mysql> update student set phoneNo=111 where id=20;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

```

```
mysql> select * from student;
```

name	id	address	marks	phoneNo
Dev	19	goa	75	0
Kalyan	20	kolkata	78	111
San	21	hyd	73	0
Pavan	22	pune	77	0
Arjun	23	angola	71	0
Sriram	24	coorg	79	0
Aniket	25	janau	70	0

```
7 rows in set (0.00 sec)
```

```
mysql> desc student;
```

Field	Type	Null	Key	Default	Extra
name	varchar(30)	YES	PRI	NULl	
id	int	NO		NULl	
address	varchar(50)	YES		NULl	
marks	int	YES		NULl	
phoneNo	int	YES		NULl	

```
5 rows in set (0.00 sec)
```

```
mysql> alter table student modify column name varchar(60);
```

```
Query OK, 0 rows affected (0.01 sec)
```

```
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> desc student;
```

Field	Type	Null	Key	Default	Extra
name	varchar(60)	YES	PRI	NULl	
id	int	NO		NULl	
address	varchar(50)	YES		NULl	
marks	int	YES		NULl	
phoneNo	int	YES		NULl	

```
5 rows in set (0.00 sec)
```

```
mysql> alter table student drop column phoneNo;
```

```
Query OK, 0 rows affected (0.02 sec)
```

```
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> select * from student;
```

name	id	address	marks
Dev	19	goa	75
Kalyan	20	kolhata	78
San	21	hyd	73
Pavan	22	pune	77
Arjun	23	ongole	71
Sriram	24	coorg	79
Aniket	25	jammu	76

```
7 rows in set (0.00 sec)
```

```
mysql> delete from student where name='San';
```

```
Query OK, 1 row affected (0.01 sec)
```

```
mysql> select * from student;
```

name	id	address	marks
Dev	19	goa	75
Kalyan	20	kolhata	78
Pavan	22	pune	77
Arjun	23	ongole	71
Sriram	24	coorg	79
Aniket	25	jammu	76

```
6 rows in set (0.00 sec)
```

```
mysql> select sum(marks) from student;
```

sum(marks)
458

```
1 row in set (0.01 sec)
```

```
mysql> select avg(marks) from student;
```

avg(marks)
75.0000

```
1 row in set (0.00 sec)
```

1 row in set (0.01 sec)

mysql> select avg(marks) from student;

avg(marks)
75.0000

1 row in set (0.00 sec)

mysql> select count(marks) from student;

count(marks)
6

1 row in set (0.00 sec)

mysql> select max(marks) from student;

max(marks)
79

1 row in set (0.00 sec)

mysql> select min(marks) from student;

min(marks)
70

1 row in set (0.00 sec)

mysql> select * from student order by marks;

name	id	address	marks
Aniket	25	jammu	70
Arjun	23	angole	71
Dev	19	goa	75
Pavan	22	pune	77
Kalyan	20	kolkata	78
Sriram	24	chennai	79

6 rows in set (0.00 sec)

```
MySQL 8.0 Command Line Client >
+-----+
| Pavan | 22 | pune | 77 |
| Kalyan | 28 | kolkata | 78 |
| Sriram | 24 | coorg | 79 |
+-----+
6 rows in set (0.00 sec)

mysql> select * from student order by marks desc;
+-----+
| name | id | address | marks |
+-----+
| Sriram | 24 | coorg | 79 |
| Kalyan | 28 | kolkata | 78 |
| Pavan | 22 | pune | 77 |
| Dev | 19 | goa | 75 |
| Arjun | 23 | ongole | 71 |
| Aniket | 25 | jammu | 70 |
+-----+
6 rows in set (0.00 sec)

mysql> select * from student where name like 'A%';
+-----+
| name | id | address | marks |
+-----+
| Arjun | 23 | ongole | 71 |
| Aniket | 25 | jammu | 70 |
+-----+
2 rows in set (0.00 sec)

mysql> select * from student where name like '%n';
+-----+
| name | id | address | marks |
+-----+
| Kalyan | 28 | kolkata | 78 |
| Pavan | 22 | pune | 77 |
| Arjun | 23 | ongole | 71 |
+-----+
3 rows in set (0.00 sec)

mysql> select * from student where name like '__v%';
+-----+
| name | id | address | marks |
+-----+
| Dev | 19 | goa | 75 |
| Pavan | 22 | pune | 77 |
+-----+
2 rows in set (0.00 sec)
```



```
MySQL 8.0 Command Line Cl  x  +  -
```

name	id	address	marks
Dev	19	goa	75
Pavan	22	pune	77

2 rows in set (0.00 sec)

```
mysql> select * from student where name like 'Sr%';
```

name	id	address	marks
Sriram	24	coorg	79

1 row in set (0.00 sec)

```
mysql> create table employ(id int not null primary key,salary int,empcode int,name varchar(60));
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> insert into employ values(1,25000,0001,'Ganesh'),(2,25500,0002,'Vignesh'),(3,26000,0003,'Dinesh'),(4,26500,0004,'Ranesh'),(5,27000,0005,'Rajesh'),(6,27500,0006,'Suresh');
Query OK, 6 rows affected (0.01 sec)
Records: 6  Duplicates: 0  Warnings: 0
```

```
mysql> select * from employ;
```

id	salary	empcode	name
1	25000	1	Ganesh
2	25500	2	Vignesh
3	26000	3	Dinesh
4	26500	4	Ranesh
5	27000	5	Rajesh
6	27500	6	Suresh

6 rows in set (0.00 sec)

```
mysql> select * from student;
```

name	id	address	marks
Dev	19	goa	75
Kalyan	20	kolkata	76
Pavan	22	pune	77
Arjun	23	ongole	71
Sriram	24	coorg	79
Aniket	25	jammu	70

6 rows in set (0.00 sec)

```
MySQL 8.0 Command Line Cl  +  x

| Sriraa | 24 | ceorg | 79 |
| Aniket | 25 | jansu | 70 |
+-----+
6 rows in set (0.00 sec)

mysql> select * from employ;
+-----+
| id | salary | empcode | name |
+-----+
| 1 | 25000 | 1 | Ganesh |
| 2 | 25500 | 2 | Vignesh |
| 3 | 26000 | 3 | Dinesh |
| 4 | 26500 | 4 | Ramesh |
| 5 | 27000 | 5 | Rajesh |
| 6 | 27500 | 6 | Suresh |
+-----+
6 rows in set (0.00 sec)

mysql> select * from student inner join employ on student.id=employ.id;
Empty set (0.00 sec)

mysql> select * from student left join employ on student.id=employ.id;
+-----+
| name | id | address | marks | id | salary | empcode | name |
+-----+
| Dev | 19 | goa | 75 | NULL | NULL | NULL | NULL |
| Kalyan | 20 | kolkata | 78 | NULL | NULL | NULL | NULL |
| Pavan | 22 | pune | 77 | NULL | NULL | NULL | NULL |
| Arjun | 23 | ongole | 71 | NULL | NULL | NULL | NULL |
| Sriraa | 24 | ceorg | 79 | NULL | NULL | NULL | NULL |
| Aniket | 25 | jansu | 70 | NULL | NULL | NULL | NULL |
+-----+
6 rows in set (0.00 sec)

mysql> select * from student right join employ on student.id=employ.id;
+-----+
| name | id | address | marks | id | salary | empcode | name |
+-----+
| NULL | NULL | NULL | NULL | 1 | 25000 | 1 | Ganesh |
| NULL | NULL | NULL | NULL | 2 | 25500 | 2 | Vignesh |
| NULL | NULL | NULL | NULL | 3 | 26000 | 3 | Dinesh |
| NULL | NULL | NULL | NULL | 4 | 26500 | 4 | Ramesh |
| NULL | NULL | NULL | NULL | 5 | 27000 | 5 | Rajesh |
| NULL | NULL | NULL | NULL | 6 | 27500 | 6 | Suresh |
+-----+
6 rows in set (0.00 sec)
```

				6	27580	6	Suresh
--	--	--	--	---	-------	---	--------

6 rows in set (0.00 sec)

mysql> select * from student cross join employ;

name	id	address	marks	id	salary	empcode	name
Aniket	25	jamu	70	1	25000	1	Ganesh
Sriram	24	coorg	79	1	25000	1	Ganesh
Arjun	23	ongole	71	1	25000	1	Ganesh
Pavan	22	pune	77	1	25000	1	Ganesh
Kalyan	20	kolkata	78	1	25000	1	Ganesh
Dev	19	goa	75	1	25000	1	Ganesh
Aniket	25	jamu	70	2	25580	2	Vignesh
Sriram	24	coorg	79	2	25580	2	Vignesh
Arjun	23	ongole	71	2	25580	2	Vignesh
Pavan	22	pune	77	2	25580	2	Vignesh
Kalyan	20	kolkata	78	2	25580	2	Vignesh
Dev	19	goa	75	2	25580	2	Vignesh
Aniket	25	jamu	70	3	26080	3	Dinesh
Sriram	24	coorg	79	3	26080	3	Dinesh
Arjun	23	ongole	71	3	26080	3	Dinesh
Pavan	22	pune	77	3	26080	3	Dinesh
Kalyan	20	kolkata	78	3	26080	3	Dinesh
Dev	19	goa	75	3	26080	3	Dinesh
Aniket	25	jamu	70	4	26580	4	Ranesh
Sriram	24	coorg	79	4	26580	4	Ranesh
Arjun	23	ongole	71	4	26580	4	Ranesh
Pavan	22	pune	77	4	26580	4	Ranesh
Kalyan	20	kolkata	78	4	26580	4	Ranesh
Dev	19	goa	75	4	26580	4	Ranesh
Aniket	25	jamu	70	5	27080	5	Rajesh
Sriram	24	coorg	79	5	27080	5	Rajesh
Arjun	23	ongole	71	5	27080	5	Rajesh
Pavan	22	pune	77	5	27080	5	Rajesh
Kalyan	20	kolkata	78	5	27080	5	Rajesh
Dev	19	goa	75	5	27080	5	Rajesh
Aniket	25	jamu	70	6	27580	6	Suresh
Sriram	24	coorg	79	6	27580	6	Suresh
Arjun	23	ongole	71	6	27580	6	Suresh
Pavan	22	pune	77	6	27580	6	Suresh
Kalyan	20	kolkata	78	6	27580	6	Suresh
Dev	19	goa	75	6	27580	6	Suresh

36 rows in set (0.00 sec)

```
mysql> show tables;
```

Tables_in_nit
customer
customers
emp
employ
employee
student

```
6 rows in set (0.02 sec)
```

```
mysql> delete customer;
```

```
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near ''  
at line 1
```

```
mysql> delete table 'customer';
```

```
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 't  
able 'customer'' at line 1
```

```
mysql> delete table customer;
```

```
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 't  
able customer' at line 1
```

```
mysql> drop table customer;
```

```
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> drop table customers;
```

```
Query OK, 0 rows affected (0.11 sec)
```

```
mysql> drop table emp;
```

```
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> drop table employee;
```

```
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> show tables;
```

Tables_in_nit
employ
student

```
2 rows in set (0.00 sec)
```

```
mysql> exit
```